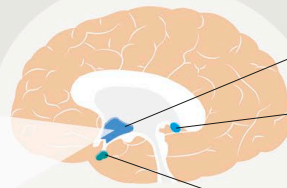




Hormone producers

Hormones are used for two types of communications. The first is between two endocrine glands, where a hormone is released to stimulate a target gland to alter the amount of hormone it is secreting. The second is between a gland and a target organ, such as the release of insulin from the pancreas prompting muscle cells to take up glucose.

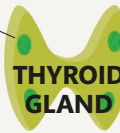


Hypothalamus links nervous system to endocrine system

Pineal gland releases melatonin in response to light levels—melatonin governs body's circadian rhythm and regulates some reproductive hormones

Controlled by the hypothalamus, pituitary gland acts as "master gland"; it secretes its own hormones that control other glands

Thyroid gland and parathyroid glands regulate metabolism, blood calcium levels, and heart rate



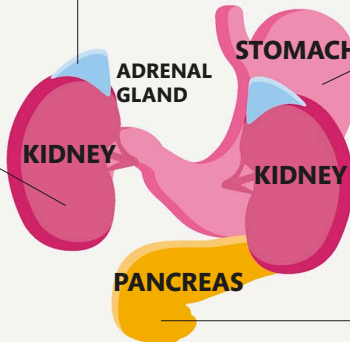
PARATHYROID GLAND

Produces cortisol (regulates metabolism, immune response, and energy conversion), aldosterone (controls blood pressure and salt balance), and adrenaline (fight-or-flight hormone)



Produces white blood cells that defend against viruses and infections

Secretes renin and angiotensin, which control blood pressure, as well as erythropoietin, which stimulates production of red blood cells



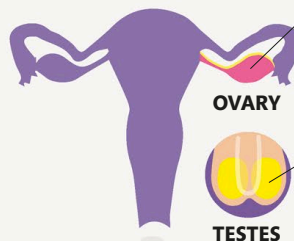
STOMACH

Releases hunger-inducing hormone ghrelin and hormone gastrin, which stimulates acid production

Secretes insulin, glucagon, and somatostatin to control blood sugar; gastrin, which stimulates stomach cells to produce acid; and a hormone that controls water secretion and absorption in intestines

Producing hormones

The endocrine system is made up of glands that are dedicated specifically to secreting hormones, as well as organs—such as the stomach—that are not glands themselves but are able to produce, store, and release hormones. Both types react to signals from the brain by increasing or decreasing the production of hormones, which then travel, via the bloodstream, to a target organ, where they lock onto specialized receptors on the surfaces of cells. This triggers a physiological change that restores homeostasis.



OVARY

Produces female reproductive hormones estrogen and progesterone, which prepare uterus for menstruation or pregnancy

TESTES

Produce testosterone, which is essential in sperm production, maintaining muscle mass and strength, libido, and bone density